Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
B		2516029

The Art of Balance: A Win-Win Strategy for Optimizing Tourist Numbers and Environmental Protection

Summary

In this study, a mathematical model based on multi-objective optimization, sensitivity analysis and sensitivity analysis was constructed to address the sustainable development of the city of Juneau, Alaska, in the midst of rapid tourism development. By analyzing the relationship between the number of tourists, economic gain, environmental pressure and community satisfaction, a set of policy recommendation for balancing economic development and environmental protection is developed. The study not only provides a scientific basis for sustainable tourism in the city of Juneau, but also provides a reference framework for the management of other over-touristed areas[1].

For Problem 1, A sustainable tourism optimization model was constructed to balance tourist numbers, revenue, and environmental impact. Using multi-objective optimization, the model maximizes economic benefits, minimizes environmental damage, and improves community satisfaction under constraints like carrying capacity and carbon emission thresholds. A revenue allocation plan directs tourism taxes to environmental protection, community projects, and long-term sustainability. The model suggests an optimal solution with around 1.13 million tourists annually, limiting cruise ships to 4 and applying a 6% consumption tax to reduce environmental stress and ensure sustainability.

For Problem 2, we adjusted the model to focus on Venice by modifying the weights and constraints to prioritize regional concerns such as historical preservation and local community well-being. The policy adjustments redirected tourists from overcrowded areas to less-visited sites, promoting sustainable tourism. Additionally, a dynamic tourist capacity strategy was introduced. The results highlight the model's flexibility, demonstrating its ability to ease congestion while supporting tourism sustainability in Venice.

For Problem 3, We developed policy recommendations for the Juneau Tourism Commission that projected tourism trends for the next decade. sensitivity modeling revealed that increasing visitor taxes, capping daily visitors, and reinvesting in infrastructure and conservation are critical to sustainable growth. Of the strategies evaluated, the "balanced development" approach proved to be the most effective, reducing environmental stress while increasing resident satisfaction. The model provides actionable recommendations for Juneau, as well as for similar destinations facing overtourism challenges.

Finally, the study effectively combines economic, environmental, and social factor s into a cohesive model, providing a scalable framework for sustainable tourism management. The model's flexibility ensures its applicability across different regions, empowering policymakers to mitigate overtourism while safeguarding long-term economic growth and environmental preservation.

Keywords: Tourism Sustainability, Multi-objective Optimization, sensitivity Analysis, Regional Priorities, Dynamic Tourist Capacity

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1 Introduction

1.1 Problem Background

Juneau, Alaska, a city with a population of approximately 30,000, has seen a dramatic rise in tourism. In 2023, it hosted 1.6 million cruise passengers, with some days accommodating over 20,000 visitors. This influx has contributed significantly to the local economy, generating an estimated $375 million annually. However, overtourism has brought pressing challenges, including environmental degradation, infrastructure overuse, and social tensions. Mendenhall Glacier, a prime attraction, has receded significantly due to climate change, exacerbated by increased tourism-related activities. The sustainability of Juneau's tourism industry is at a crossroads, necessitating a strategic balance between economic growth and resource preservation[2].



(a) Mendenhall Glacier before melting



(b) Mendenhall Glacier after melting

Figure 1: Mendenhall Glacier before and after melting

(a) **Mendenhall Glacier before melting**: Shows the glacier as it was when it was not significantly affected by climate change, with a large glacier extent and a relatively stable environment.

(b) **Mendenhall Glacier after melting**: Shows the current state of the glacier after significant retreat, with the glacier's extent significantly reduced and the surrounding ecosystem changed, highlighting the impacts of climate change and over-tourism.

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- Develop a mathematical model for a sustainable tourism industry in Juneau, Alaska. The model should balance economic growth, environmental preservation, and local community well-being by optimizing factors such as visitor numbers, overall revenue, and resource allocation. Constraints such as environmental capacity, community satisfaction, and infrastructure limitations should be integrated into the model.
- Evaluate how the proposed model can be adapted to other tourist destinations affected by overtourism. Identify how location-specific factors, such as environmental sensitivity and tourist behavior, influence the effectiveness of various measures, and propose strategies for promoting under-visited attractions to achieve

a more balanced tourism distribution.

- Provide actionable recommendations to the Juneau Tourist Council. Prepare a one-page memo outlining model predictions, the impacts of different measures, and policy advice to optimize tourism outcomes while addressing economic, environmental, and social priorities.

1.3 Our Work

➢ In order to solve the mentioned problem, our work can be illustrated by the following flowchart.

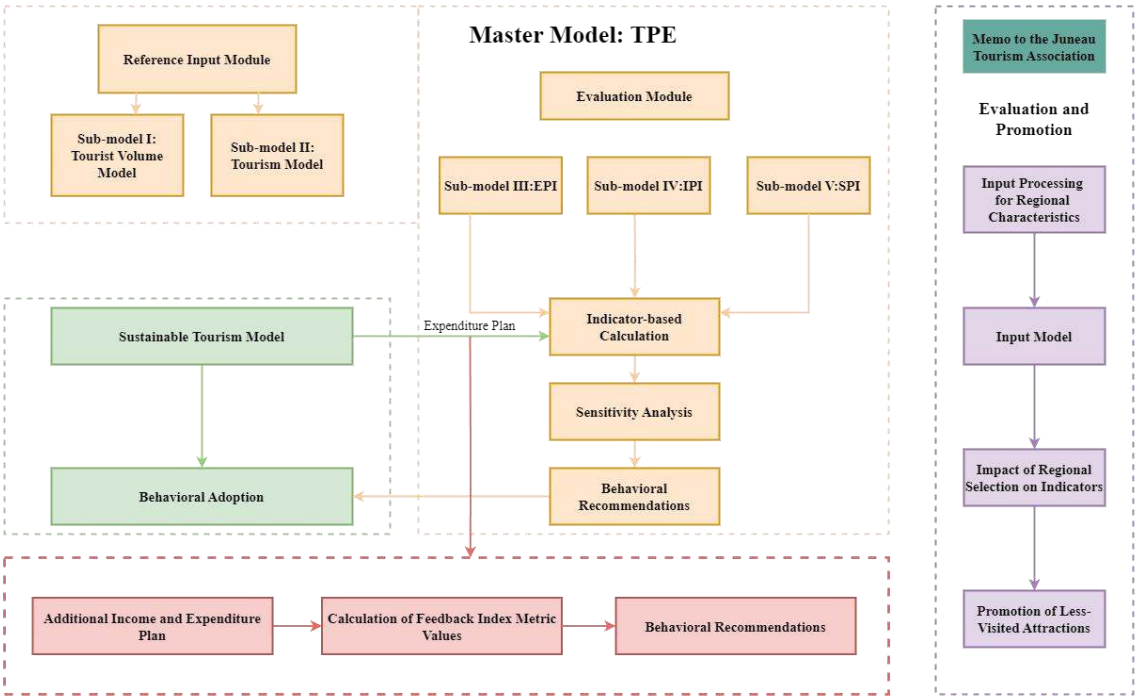


Figure 2: The Framework of Our Work

2 Assumptions and Justifications

To simplify the problem, we make the following basic assumptions, each of which is properly justified.

✧ **Assumption 1: Tourist numbers and revenue are linearly related.**

Justification: It is assumed that the number of tourists is linearly related to tourism revenue, meaning that each additional tourist increases revenue by a fixed amount. This assumption is based on the stability of tourist spending behavior, where the average expenditure per tourist does not significantly change with variations in tourist numbers.

✧ **Assumption 2: Environmental costs are non-linearly related to tourist numbers.**

Justification: It is assumed that environmental costs (e.g., carbon emissions) are proportional to the square of tourist numbers, meaning that as the number of tourists increases, environmental costs grow at an accelerating rate. This assumption reflects the

finite nature of environmental resources, where environmental pressure increases significantly once tourist numbers exceed a certain threshold.

✧ **Assumption 3: Social costs are related to tourist numbers.**

Justification: It is assumed that social costs (e.g., a decline in residents' quality of life) have an exponential relationship with tourist numbers. Once tourist numbers exceed a certain threshold, social costs rise sharply. This assumption is based on residents' sensitivity to overtourism, where excessive tourist numbers significantly impact residents' quality of life.

✧ **Assumption 4: Infrastructure has limited carrying capacity.**

Justification: It is assumed that the infrastructure in Juneau (e.g., transportation, accommodation) has a fixed carrying capacity. Once this capacity is exceeded, tourist experiences and residents' quality of life deteriorate significantly. This assumption reflects the physical limits of infrastructure, where the capacity is finite.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description	Unit
$N(m, y)$	Number of visitors in month m and year y	People
$R(m, y)$	Tourism revenue in month m and year y	USD
R_{cruise}	Revenue from cruise ships	USD
R_{air}	Revenue from air travel	USD
R_{road}	Revenue from road and ferry travel	USD
$R_{visitor}$	Revenue from food, accommodation, and activities	USD
$E(m, y)$	The operating cost in month m and year y	USD
$EPI(t)$	Environmental Pressure Index	Unitless
$IPI_{total}(t)$	Total Infrastructure Pressure Index at time t	Unitless
$SPI(t)$	Social Pressure Index at time t	Unitless
c_{res} /c_{tour}	Carbon emissions by residents/tourists	Tons CO_2/day
w_{res} /w_{tour}	Water usage by residents/tourists	m^3/day
g_{res} /g_{tour}	Waste production by residents/tourists	kg/day

4 Model Preparation

4.1 The Data

4.1.1 Data Collection

The data used in this study includes historical population data, tourist statistics, energy demand, transportation metrics, water usage, and environmental impact indicators. Table 2 summarizes the main data sources:

Table 2: Data source collation

Database Names	Database Websites Data	Type
PadMapper	https://www.padmapper.com/	Housing and rental costs
Numbeo	https://www.numbeo.com/	Income and satisfaction data
Alaska Department of Labor	https://labor.alaska.gov/	Local CPI and labor statistics
Data USA	https://datausa.io/	Transportation and commute time data
Neilsberg	https://www.neilsberg.com/	Resident population statistics
Travel and Tour World	https://www.travelandtourworld.com/	Tourist arrival statistics
Alaska Energy Authority	https://www.akenergyauthority.org/	Energy demand and consumption
Alaska Energy Authority	https://dot.alaska.gov/	Road usage data
Travel Juneau	https://www.traveljuneau.com/	Tourism facilities and attractions
Alaska EPA	https://www.epa.gov/	Environmental data (carbon, waste, water)

4.1.2 Data Inference

To address data gaps, our team used reliable sources to derive and adjust estimates. We estimated CPI at 2.5%-3.5% based on Juneau's unique costs. Electricity demand was calculated as 24 kWh per person daily, while waste generation totaled 46.07 tons daily, following high-income country standards. Adjusted for hydroelectric energy use, carbon emissions were estimated at 0.044 tons per person daily. Water consumption matched the U.S. average at 0.35 m^3 per person daily. Road usage was estimated at 214,000-229,000 vehicle miles daily, and building facility demand was set at 765 square feet per person. These estimates provide robust data for our model.The derivation of the specific model input data is shown below:

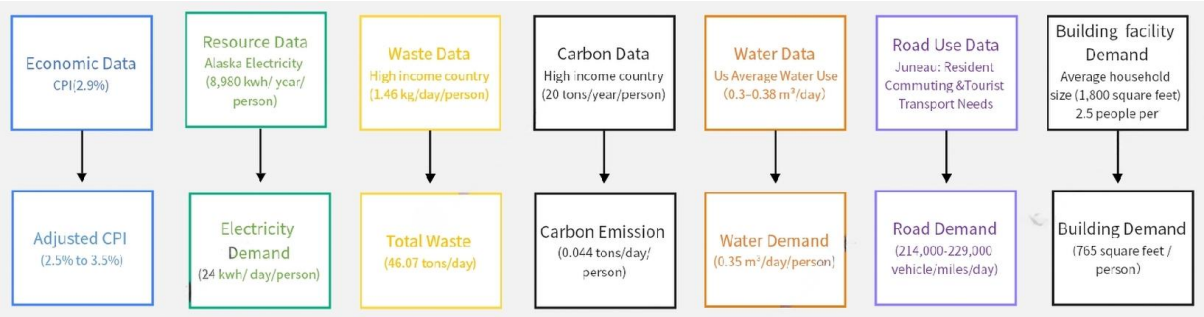


Figure 3: Data derivation for model inputs

5 Models and Solution of Problem 1

5.1 Overview of Problem 1

To address sustainable tourism in Juneau, we aim to build a comprehensive model that balances economic benefits, environmental protection, infrastructure resilience, and social satisfaction. This involves analyzing the dynamic relationships between visitor numbers, revenue, and their impacts. The goal is to optimize these factors within defined constraints and propose a sustainable development strategy for tourism[3].

5.2 Sub-Models Construction

5.2.1 Visitor Flow Model

In order to better simulate the seasonal changes in the number of tourists in Juneau, we constructed a visitior flow model to accurately capture the dynamic changes in the number of tourists by combining seasonal fluctuation factors through the compound annual growth rate (CAGR),The CAGR is calculated as follows:

$$CAGR = \left(\frac{P_{final}}{P_{initial}}\right)^{\frac{1}{t_{final}-t_{initial}}} - 1 \quad (1)$$

The Visitor Flow Modeling formulas are as follows::

$$N(m,y) = \left(A_0 \cdot (1+r)^{y-y_0} \cdot sin\left(B \cdot (m-C)\right) + D_0 \cdot (1+r)^{y-y_0}\right) + \epsilon(m,y) \quad (2)$$

where $N(m,y)$ is the number of visitors in month m and year y; A_0 is the magnitude of fluctuations in the number of visitors in the base year; r is the average compounded growth rate (CAGR), which is about 5.3%; B is the frequency of fluctuations in the annual cycle; C is the peak month offset; D_0 is the average number of visitors in the base year; and $\epsilon(m,y)$ is the external stochastic perturbation, described by the $N(0,\sigma^2)$ -normal distribution, for example, the effect of weather or unexpected events.

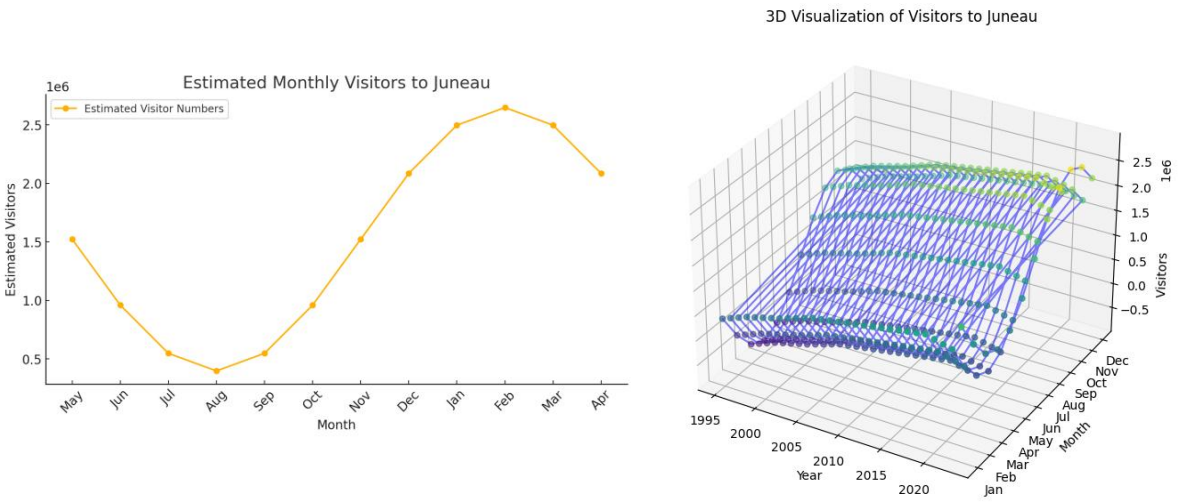


Figure 4: Estimated monthly and long-term trends in visitors to Juneau

The model not only captures the long-term growth trend of tourist numbers but also visualizes the seasonal fluctuation patterns throughout the year by introducing parameters such as fluctuation frequency and month offset. Notably, tourist numbers increase significantly during the summer peak season and decrease sharply during the winter off-season. The left graph illustrates the seasonal variations in visitor numbers within a single year, while the right graph highlights the combined effects of long-term growth and seasonal fluctuations over multiple years. Together, these visualizations validate the robustness and practical applicability of the Visitor Flow Model.

5.2.2 Economic Revenue Model

The economic revenue model evaluates tourism revenue in Juneau by dividing income into cruise-related and other categories. Monthly revenue is calculated as follows:

$$R(m,y) = N(m,y) \cdot (0.65 \cdot R_{cruise} + 0.32 \cdot R_{air} + 0.03 \cdot R_{road} + R_{visitor} - E(m,y) \quad (3)$$

Where R_{cruise} represents the economic benefits generated by cruise ships, such as docking fees and port charges paid by the cruise lines;R_{air} denotes the average spending by air travelers; R_{road} refers to the average spending by road or ferry travelers; $R_{visitor}$ indicates the average spending by tourists on food, accommodation, transportation, and other activities;$E(m,y)$is the operating cost, dynamically calculated based on the number of visitors and resource consumption; and $N(m,y)$ is derived from the visitor flow model. All the above expenditures include a 5% tax.

5.2.3 Environmental Pressure Model

In order to analyze the comprehensive impact of tourism activities on the environment, the Environmental Pressure Integrated Index (EPI) was constructed, with total carbon emissions, total water use, total waste production, and natural habitat overload as the main indicators. The Environmental Pressure Index (EPI) is defined as follows:

$$EPI(t) = w_C \cdot C(t) + w_W \cdot W(t) + w_G \cdot G(t) \quad (4)$$

Where C is the total carbon emissions; W is the total water usage; G is the total waste production; and w_C,w_W,w_G are the weight coefficients.

Continuing from where the Environmental Pressure Index (EPI) is introduced, the detailed components of the model are expressed below. Each indicator reflects a critical environmental factor influenced by tourism activities:

(1) Total Carbon Emissions

Carbon emissions are derived from both residents and tourists. The total carbon emissions can be calculated as:

$$C(t) = N_{res} \cdot c_{res} + N(t) \cdot c_{tour} \quad (5)$$

Where N_{res} is the local resident population; N is the total number of tourists (from the visitor number model); c_{res} is the per capita carbon emissions of local residents (kg/day); and c_{tour} is the per capita carbon emissions of tourists (kg/day).

(2) Total water consumption

$$W(t) = N_{res} \cdot w_{res} + N(t) \cdot w_{tour} \quad (6)$$

Where w_{res} is the daily water consumption per resident (liter/day); w_{tour} is the daily water consumption per visitor (liter/day).

(3) Total waste generation

$$G(t) = N_{res} \cdot g_{res} + N(t) \cdot g_{tour} \quad (7)$$

Where g_{res} is the amount of waste generated per resident per day (kg/day); g_{tour} is the amount of garbage generated per tourist per day (kg/day).

Combining the number of tourists and environmental pressure, the optimization objective is to minimize EPI while meeting certain tourist growth demands:

$$min\ EPI \quad (8)$$

Figure 3 demonstrates how adjusting the weights of key environmental factors—carbon emissions, water usage, and waste generation—affects the overall EPI. The left grap hhighlights the contributions of each factor to the EPI under different weight combinations, while the right graph offers a 3D visualization of how these weights interact, helping to understand the cumulative environmental pressure as the number of tourists increases.

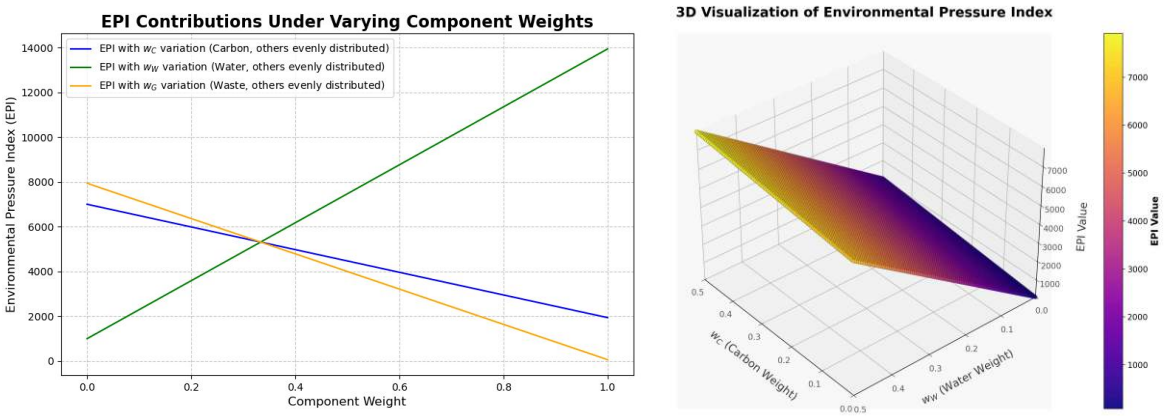


Figure 5: EPI Contributions and 3D Visualization Under Varying Component Weights

5.2.4 Infrastructure Pressure Model

To analyze the comprehensive pressure exerted on infrastructure by tourism activities, the Infrastructure Pressure Index (IPI) is constructed, encompassing electricity demand, road usage demand, and building facility demand as the main indicators. The Infrastructure Pressure Index (IPI) is defined as follows:

$$IPI_{total}(t) = w_E \cdot D_E(t) + w_R \cdot D_R(t) + w_B \cdot D_B(t) \quad (9)$$

Where $D_E(t)$ is the demand for electricity;$D_R(t)$ is the demand for roads; and $D_B(t)$ is the demand for building facilities;

(1) Electricity demand

Electricity demand is influenced by both local residents and tourists. The total electricity demand can be calculated as:

$$D_E(m) = P_r \cdot UE_r + P(t) \cdot UE_t \quad (10)$$

where P_r is the total number of local residents; $P_t(t)$ is the total number of tourists at time; UE_r is the month per capita electricity consumption of residents (kWh/month); and UE_t is the month per capita electricity consumption of tourists (kWh/month).

(2) Road demand

Road demand reflects the demand of residents and tourists to use the transportation system, and the expression is:

$$D_R(t) = P_r \cdot UR_r + P(t) \cdot UR_t \quad (11)$$

Where UR_r is the per capita road use coefficient of residents; UR_t is the per capita road use coefficient of tourists.

(3) Demand for building facilities

Building facilities demand measures the demand of residents and tourists for the use of public buildings and facilities, and the expression is:

$$D_B(t) = P_r \cdot UB_r + P(t) \cdot UB_t \quad (12)$$

The optimization objective is to minimize $IPI_{total}(t)$ while meeting infrastructure capacity constraints and achieving tourism growth needs:

$$min\ IPI_{total}(t) \quad (13)$$

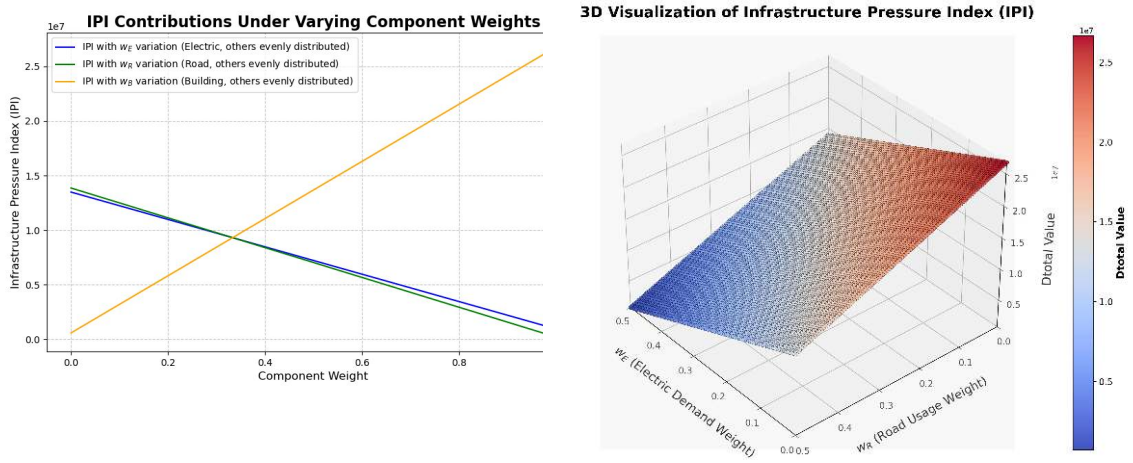


Figure 6: IPI Contributions and 3D Visualization Under Varying Component Weights

The left graph shows how the Infrastructure Pressure Index (IPI) varies with different component weights (Electric, Road, Building). As the weight of Electric and Road decreases, IPI decreases, while increasing Building weight raises IPI. The right graph presents a 3D view of how Electric and Road weights affect the IPI, with color indicating pressure levels.

5.2.5 Social Pressure Model

To analyze the comprehensive pressure exerted on society by urban development and tourism activities, the Social Pressure Index (SPI) is constructed, encompassing housing pressure, cost of living pressure, traffic pressure, and resident satisfaction as the main indicators. The Social Pressure Index (SPI) is defined as follows:

$$SPI(t) = w_H \cdot H(t) + w_C \cdot C(t) + w_T \cdot T(t) + w_S \cdot S(t) \tag{14}$$

Where $H(t)$ represents housing pressure; $C(t)$ represents cost-of-living pressure, indicating the economic pressure caused by daily expenses; $T(t)$ represents transportation pressure, measuring the impact of commuting and traffic congestion on residents; $S(t)$ represents resident satisfaction, an inverse indicator reflecting residents' overall well-being; andw_H,w_C,$w_T$and w_S are weighting coefficients.

(1) Housing Pressure

$$H(t) = \frac{\text{Average Housing Cost}}{\text{Average Income}} \tag{15}$$

(2) Cost of Living Pressure

Expressed directly using the local price index (CPI) or Price Level:

$$C(t) = \text{CPI or Price Level} \tag{16}$$

(3) Traffic Pressure

Traffic pressure is expressed using Average Commute Time or Traffic Congestion Index:

$$T(t) = \text{Average Commute Time} \tag{17}$$

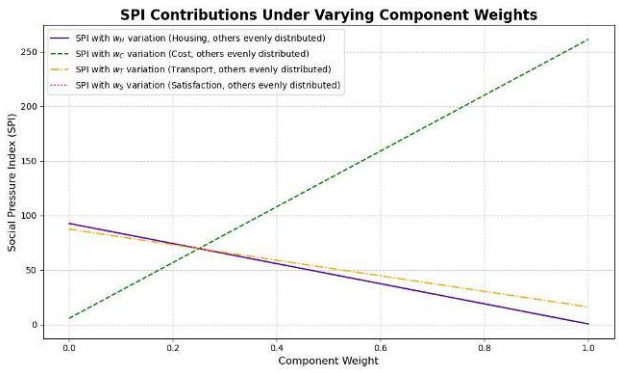
(4) Resident Satisfaction

Resident satisfaction is an inverse indicator of social pressure and therefore needs to be treated in reverse:

$$S(t) = \frac{\mathrm{S_{max}} - \mathrm{S_{current}}}{\mathrm{S_{max}}} \tag{18}$$

The balance between social development and the quality of life of residents is achieved through the following objectives:

$$min\ SPI(t) \tag{19}$$



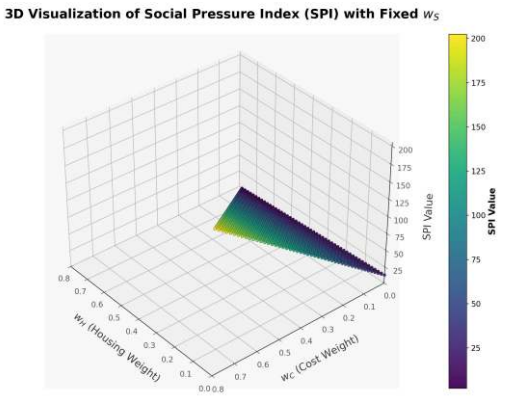


Figure 7: SPI Contributions and 3D Visualization Under Varying Component Weights

The left graph illustrates the Social Pressure Index (SPI) contributions with varying component weights (Housing, Cost, Transport, and Satisfaction). As the weight for Housing decreases and Transport increases, the SPI value increases. The right graph provides a 3D visualization of the SPI with fixed Satisfaction weight , showing how Housing and Cost weights affect the SPI value, with the color gradient representing the pressure levels.

5.3 Optimization Objectives and Constraints

The final optimization objective integrates all sub-models to achieve a balance between maximizing economic benefits(R(m,y)) and minimizing pressures (EPI, IPI, SPI)[4].

$$Maximize\ Z(m,y) = \gamma R(m,y) - \lambda_1 EPI - \lambda_2 IPI - \lambda_3 SPI \quad (20)$$

where $R(m,y)$ denotes tourism revenue; -EPI denotes environmental pressure; -IPI denotes infrastructure pressure; -SPI denotes social pressure; and λ_1, λ_2, and λ_3 are penalty coefficients used to balance the relationship between revenue maximization and various types of constraints.

1) Incorporating Compensation Mechanism

Recognizing that pressures may approach their respective thresholds, a compensation mechanism is introduced to:

(1) Mitigate the negative impact of each pressure (EPI,IPI,SPI) when close to its threshold.

(2) Proportionally distribute compensation based on the relative importance (λ1,λ2,λ3) of each pressure.

To address these pressures effectively and promote sustainable development, the compensation mechanism incorporates an income allocation plan aimed at:

- **Environmental Protection Programs:** Funding habitat restoration initiatives and carbon offset projects to reduce the environmental impact of tourism.
- **Infrastructure Upgrades:** Expanding road capacity and improving the electricity network to enhance infrastructure resilience.
- **Community Improvement Projects:** Supporting affordable housing and public transportation development to improve residents' quality of life.

The compensation terms are designed as follows:

For EPI：

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} EPI \times 5\%$$

For IPI：

$$\frac{\lambda_2}{\lambda_1 + \lambda_2} IPI \times 5\%$$

For SPI：

$$\frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} SPI \times 5\%$$

2) Final Objective Function (Including Compensation Terms)

The compensation terms are incorporated into the original formula (18) to address the penalties and ensure a balanced optimization:

$$\begin{aligned} Z(m,y) \quad &= \gamma R(m,y) - \lambda_1 EPI - \lambda_2 LPI - \lambda_3 SPI + \frac{\lambda_1}{\lambda_1+\lambda_2} EPI \times 5\% \\ &+ \frac{\lambda_2}{\lambda_1+\lambda_2} IPI \times 5\% + \frac{\lambda_3}{\lambda_1+\lambda_2+\lambda_3} SPI \times 5\% \end{aligned} \tag{21}$$

Constraints:

The optimization is subject to the following constraints to ensure the feasibility and sustainability of the model:

(1) Environmental Constraint:

$$E(t) \leq E_{max} \tag{22}$$

- $E(t)$:Environmental pressure at time t.
- E_{max}:Maximum allowable environmental pressure.

(2) Infrastructure Constraint:

$$I(t) \leq I_{max} \tag{23}$$

- $I(t)$:Infrastructure pressure at time t.
- I_{max}: Maximum allowable infrastructure pressure.

(3) Social Constraint:

$$S(t) \leq S_{max} \tag{24}$$

- $S(t)$:Social pressure at time t.
- S_{max}: Maximum allowable social pressure.

(4) Tourist Number Constraint:

$$N_{min} \leq N(t) \leq N_{max} \tag{25}$$

- $N(t)$:Number of visitors at time t.
- N_{min}:Minimum acceptable visitor count.
- N_{max}: Maximum allowable visitor count.

5.4 Model Solving

To solve the comprehensive model for sustainable tourism in Juneau, the data-driven approach integrates the sub-models (Visitor Flow, Economic Revenue, Environmental Pressure, Infrastructure Pressure, and Social Pressure) with the specific input data from real-world sources. This section outlines the solution process using the provided data to validate and optimize the model.

5.4.1 Input Data Integration

The model leverages real-world data to parameterize the sub-models. The detailed input data for SPI, IPI, and EPI are summarized in the following Table 3,4,5:

(1) Social Pressure Index (SPI)

Table 3: Input Data Integration(SPI)

Index	Input Data	Source
Monthly Rent	$1,750	PadMapper
Average Monthly Income	$4,159.50	Numbeo
Local CPI	261.34	Alaska Department of Labor
Average Commute Time	16 minutes	Data USA
Satisfaction Score	7	Numbeo

(2) Infrastructure Pressure Index (IPI)

Table 4: Input Data Integration (IPI)

Index	Input Data	Source
Resident Population	31,555	Neilsberg
Tourist Numbers	8,300	Travel and Tour World
Per Capita Daily Electricity Demand	24 kWh/person	Alaska Energy Authority
Total Road Usage	214,320 vehicle miles/day	Alaska Department of Transportation
Building Facility Demand	26,754,075 ft^2/day	Travel Juneau, Local Statistics

(3) Environmental Pressure Index (EPI)[5]

Table 5: Input Data Integration (EPI)

Index	Input Data	Source
Per Capita Carbon Emissions (Residents)	0.044 tons CO_2/day	EPA
Per Capita Carbon Emissions (Tourists)	0.066 tons CO_2/day	EPA
Daily Water Usage per Person	0.35 m^3/day	Alaska Environmental Protection
Waste Production per Person	0.00146 tons/day	Alaska Environmental Protection

This table summarizes the critical input parameters and their sources for calibrating the sub-models. These data points ensure the robustness and applicability of the integrated framework to solve the optimization objectives.

5.4.2 Visualization and Interpretation

This section provides a comprehensive visualization of the seasonal trends in visitor numbers and the corresponding environmental pressure. The combination of a polar plot and a 3D surface plot reveals both cyclical and temporal dynamics in tourism activities, offering critical insights for sustainable tourism planning[6].

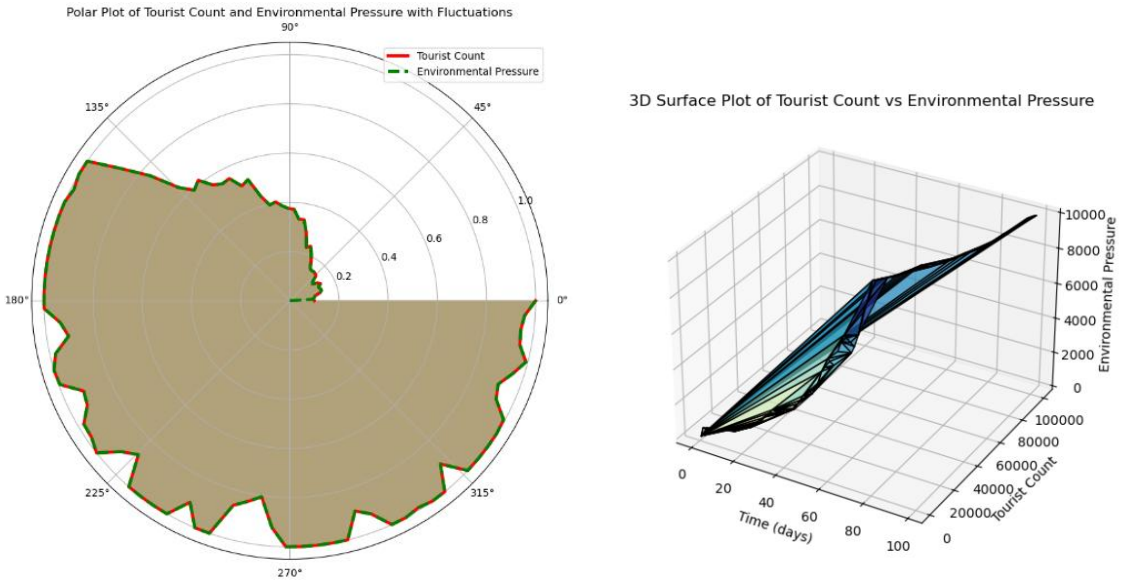


Figure 8: Dynamic Trends of Visitor Numbers and Environmental Pressure

1) **Polar Plot:**The polar plot highlights the cyclical nature of tourism dynamics by illustrating the monthly variations in visitor numbers and their corresponding environmental pressure. Peaks in the plot represent the summer tourist season when both visitor numbers and environmental pressure reach their maximum, while troughs correspond to the winter off-season.

2) **3D Surface Plot:** The 3D surface plot provides a detailed temporal view, capturing how visitor growth over time correlates with increasing environmental pressure. The gradient indicates the intensification of environmental stress during peak tourist periods, underscoring the need for intervention strategies such as visitor caps or the promotion of eco-tourism.

This boxplot visualizes the distribution of three key metrics: Tourist Count, Environmental Pressure, and Infrastructure Burden. It highlights the variability, central tendency (median), and potential outliers for each indicator. The chart provides insights into how these factors fluctuate and their relative scales.

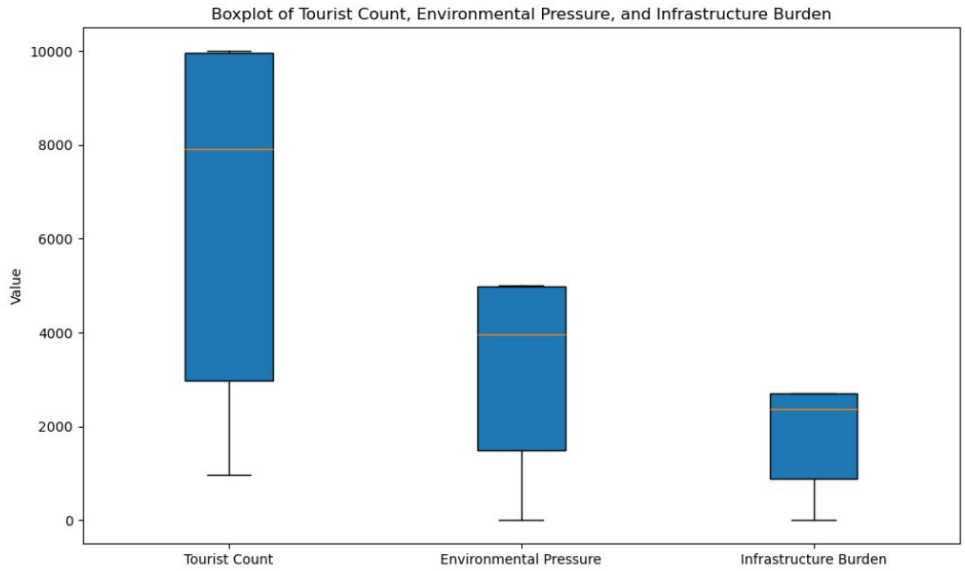


Figure 9: Comparison of Variability in Tourist Count, Environmental Pressure, and Infrastructure Burden

5.4.3 Key Results and Implications

(1) Visitor Growth:

Simulations reveal a steady increase in visitor numbers within defined constraints. However, during peak seasons, visitor numbers approach the maximum capacity, signaling the importance of proactive management strategies.

(2) Environmental Pressure:

During peak tourism periods, environmental pressure ($E(t)$) nears its threshold (E_{max}). This emphasizes the need for measures such as promoting eco-tourism or imposing visitor caps during peak seasons.

(3) Infrastructure Burden:

Infrastructure pressure ($I(t)$) remains manageable but increases proportionally with visitor numbers. Strategic investments in infrastructure—such as road expansion and energy grid upgrades—are critical for long-term sustainability.

(4) Social Pressure:

Social Pressure Index (SPI) analysis reveals that housing and transportation pressures are under control. However, rising living costs (CPI) demand attention, particularly as visitor numbers grow.

5.4.4 Sensitivity Analysis

The sensitivity analysis heatmap below illustrates the correlation strength between key input variables and output indicators in the tourism system. Red indicates a positive correlation, while blue indicates a negative correlation.

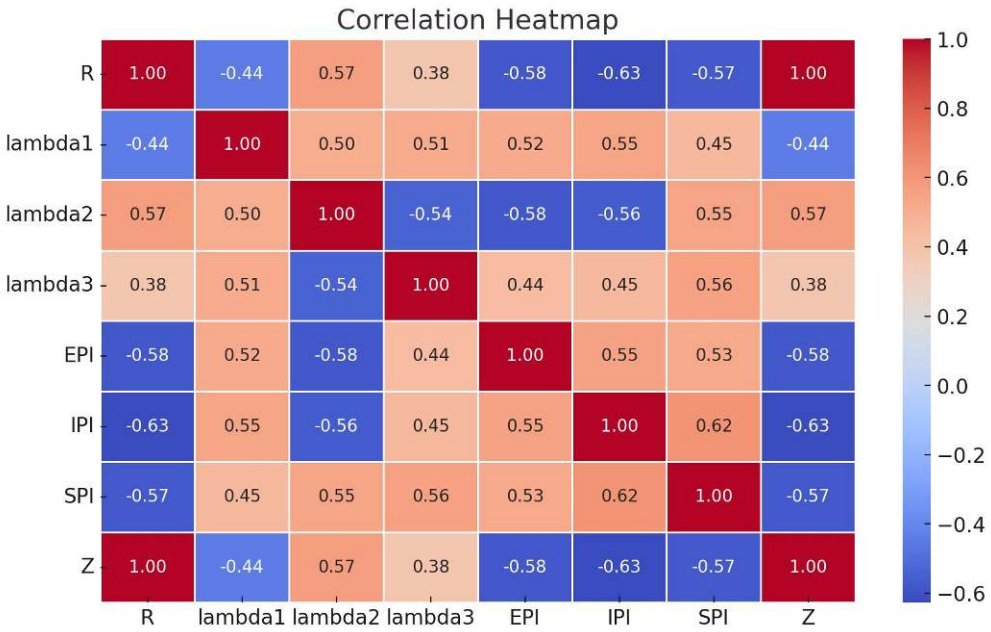


Figure 10: Correlation Heatmap

It can be observed that revenue is positively correlated with tourist numbers, whereas EPI, IPI, and SPI are negatively correlated with tourist numbers. The model's solution determines how to balance tourist numbers with resident income, environmental protection, infrastructure sustainability, and resident satisfaction. By adjusting penalty coefficients and , we predict the balance that maximizes revenue while considering other factors.

As tourist numbers decrease, indicators such as overall carbon emissions, water usage, waste generation, electricity demand, road demand, building facility demand, housing, living costs, and traffic also decrease. This benefits environmental and infrastructure protection while improving housing, living costs, and traffic. However, this reduction in tourists results in a corresponding decline in resident income. Resident satisfaction shows an optimal range: too few tourists harm income, while too many tourists disrupt the balance, ultimately lowering satisfaction.

The model determines that approximately 1,132,679 tourists optimize the balance for the local area. The model also highlights seasonal variations: summer sees the highest tourist influx, placing pressure on infrastructure and environmental systems, while winter sees fewer tourists. The impact of cruise ships on resident satisfaction is significant, while spring and autumn exhibit improved conditions. Whale watching and tropical rainforest experiences are highly satisfactory, suggesting these resources can be further developed.

To address these insights, we recommend:

- Developing tourism resources for spring and autumn to attract more visitors during these seasons.
- Expanding air and road transportation infrastructure while limiting cruise ships to four vessels.
- Implementing a 6% consumption tax and restricting visitor numbers.
- Introducing higher ticket prices and promoting whale watching and rainforest tours.

For the second destination, Venice, this model can be adapted with minor modifications to address specific challenges. Similar strategies can be employed, such as reinvesting tourism revenue into environmental protection and local welfare to prevent resident outmigration. Raising transportation prices, diversifying tourism offerings, avoiding overreliance on single tourist attractions, and implementing higher taxation can ensure sustainable tourism and improved resident satisfaction in Venice.

6 Models and Solution of Problem 2

6.1.1 Background and Objectives

The rapid growth of tourism has brought significant economic benefits to many world-famous destinations but has also led to overtourism challenges, imposing immense pressure on the local environment, infrastructure, and social systems. Venice is a prime example of this phenomenon, hosting millions of tourists annually, with notable environmental and social consequences. Against this backdrop, we aim to adapt and optimize the existing tourism model for Venice to promote sustainable tourism development.

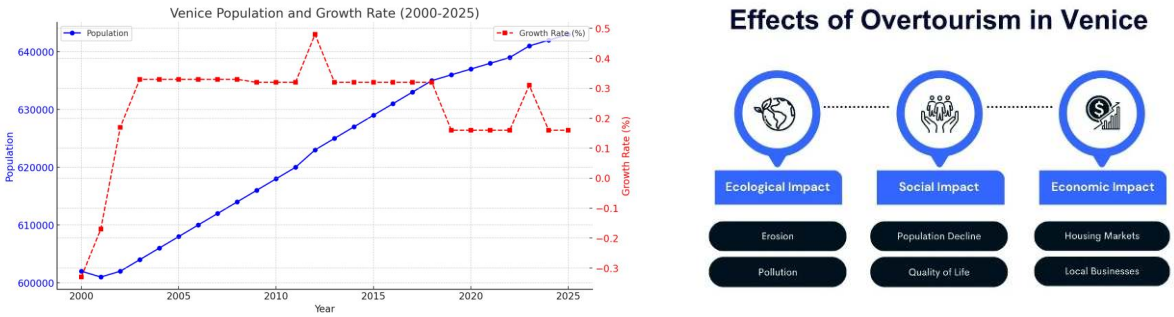


Figure 11: Population and Growth Trends in Venice

Figure 9 below illustrates the population growth and annual growth rate in Venice from 2000 to 2025. The blue line represents the steady increase in Venice's population over time, while the red dashed line highlights the fluctuating growth rate percentage.

Our work is guided by the following objectives:

- Model Adaptation: Modify the existing sub-models, including the Visitor Flow Model, Economic Revenue Model, Environmental Pressure Index (EPI), Infrastructure Pressure Index (IPI), and Social Pressure Index (SPI), to suit Venice's specific circumstances.
- Optimization Strategies: Analyze the current tourism pressures in Venice using the adjusted models and propose strategies to reduce visitor overcrowding and optimize resource allocation.
- Promotion of Less-Visited Attractions: Adjust visitor distribution to reduce pressure on popular sites and increase the attractiveness of less-visited areas.

The proposed adjustments and strategies aim to address Venice’s overtourism challenges by leveraging the existing models with tailored modifications. Below are the reasons for each proposed adjustment:

- Visitor Flow Model

1. **Input Adjustment:** Incorporating Venice's 4.51% annual tourism growth rate ensures the model accurately reflects the increasing number of visitors. This adjustment captures the long-term trend and provides a realistic projection of visitor numbers.

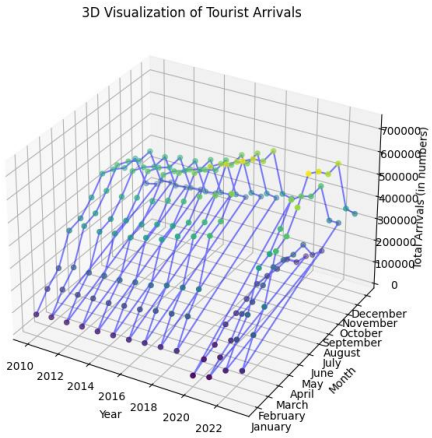


Figure 12: 3D Visualization of Tourist Arrivals(Venice)

2. **Seasonal Factors:** Including seasonal peaks, off-season variations, and government restrictions helps capture the cyclical nature of tourism. It also enables the identification of specific periods that are most congested, allowing for targeted interventions.
3. **Optimization:** Encouraging visits during less crowded periods redistributes tourism demand, reducing the burden on popular attractions and mitigating the strain on resources.

✧ Economic Revenue Model

1. **Simplification:** As Venice heavily relies on waterways for transportation, unifying transportation income provides a clearer and more manageable approach to revenue calculation. This simplification ensures consistency in the model outputs.
2. **Revenue Allocation:**Direct reinvestment of entry fees and taxes into environmental and infrastructure initiatives creates a feedback loop. This ensures that the revenue generated from tourism directly addresses the negative impacts caused by overtourism, such as environmental degradation and overburdened infrastructure.

✧ Environmental Pressure (EPI) and Infrastructure Pressure (IPI)

1. **Local Data Integration:** Adjusting variables such as carbon emissions, water usage, and building demands to Venice-specific data ensures that the model accurately reflects the city's unique environmental and infrastructural challenges.
2. **Feedback Loop:**Including revenue reinvestment mechanisms in the model demonstrates how tourism income can be effectively used to restore the environment and upgrade infrastructure, making the tourism system more sustainable.

✧ Social Pressure (SPI)

1. **Housing and Transportation Costs :** Venice's high cost of living and transportation challenges due to its historic layout are critical pressures on residents. Incorporating

these factors ensures that the model addresses the socio-economic impact of overtourism on the local community.

2. **Adjustments:** Reflecting issues like residential displacement and commuter congestion highlights the specific social challenges caused by tourism, ensuring that solutions prioritize resident well-being.

Based on model adjustments, the following strategies are recommended to address Venice's overtourism issues:

❖ Encouraging Off-Peak Visits

① Reason: Redistributing visitors across off-peak periods alleviates the pressure on popular sites during the summer, ensuring more balanced resource utilization throughout the year.

❖ Raising Entry Fees

② Reason: Higher fees during peak seasons deter excessive visitors while generating additional revenue. This revenue can fund environmental restoration and infrastructure improvements, making tourism more sustainable.

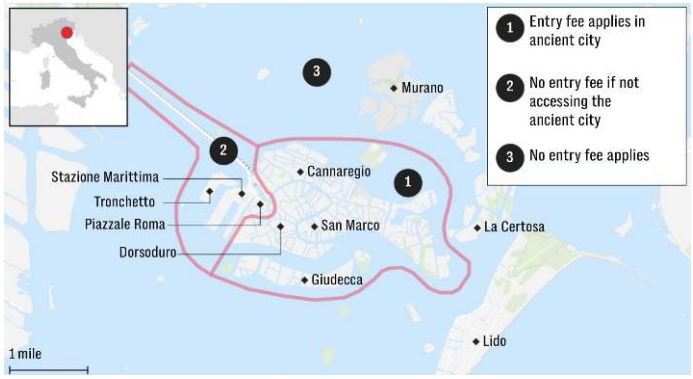


Figure 13: Entry Fee Zones in Venice and Their Impact on Visitor Regulation

❖ Capping Visitor Numbers

③ Reason: Limiting daily and annual visitor numbers prevents exceeding the city’s environmental and infrastructural capacity, protecting Venice’s long-term viability as a tourist destination.

❖ Promoting Less-Visited Attractions

④ Reason: By redistributing visitors to underutilized attractions, the pressure on iconic sites like San Marco is reduced, ensuring a more even distribution of tourists while maintaining economic benefits.

7 Model Evaluation

7.1 Strengths

1) **Holistic Integration:** The model integrates economic, environmental, infrastructure, and social factors to provide a comprehensive framework for sustainable tourism management.

2) **Adaptability:** By adjusting parameters and constraints, the model can be tailored to address the unique needs of different regions (e.g., Venice).
3) **Data-Driven Approach:** Validated with real-world data, the model ensures credibility and optimizes key variables through sensitivity analysis.

7.2 Weaknesses

1) **Limited Dynamic Adaptability:** The model lacks a real-time data update mechanism, making it less responsive to sudden events (e.g., extreme weather or policy changes) that dynamically impact tourism systems.
2) **Data Dependency:** Some parameters rely on estimates or generalized data, which may introduce uncertainties in scenarios with atypical conditions.

8 Conclusion

This study presents a robust and adaptable framework for managing sustainable tourism in Juneau, Alaska, and other overtouristed destinations. By integrating economic, environmental, infrastructure, and social factors into a multi-objective optimization model, we balance tourism-driven revenue with critical pressures such as environmental degradation, infrastructure strain, and resident dissatisfaction. Key findings highlight the importance of **capping annual tourist numbers** (e.g., 1.3 million for Juneau), **imposing visitor fees**, and **diversifying tourism activities** to alleviate seasonal overcrowding.

The model's flexibility is demonstrated through its successful adaptation to Venice, where strategies like redistributing tourist flows and prioritizing cultural preservation align with local priorities. While the framework provides actionable insights, future work could enhance its **dynamic adaptability** by incorporating real-time data updates and refining stakeholder conflict resolution mechanisms.

Ultimately, this research underscores that sustainable tourism is not a zero-sum game. By reinvesting revenue into conservation, infrastructure, and community programs, destinations like Juneau can achieve long-term economic vitality while safeguarding natural and cultural resources—a win-win strategy for residents, tourists, and ecosystems alike.

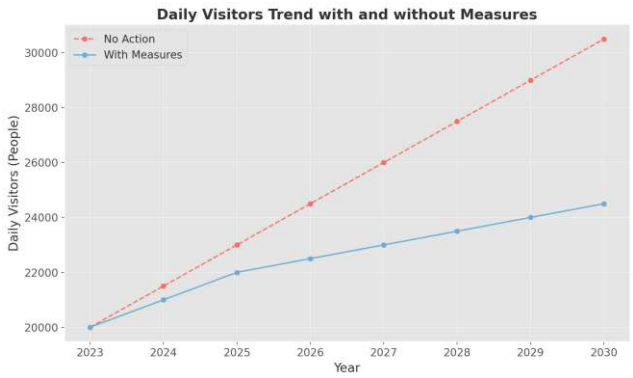


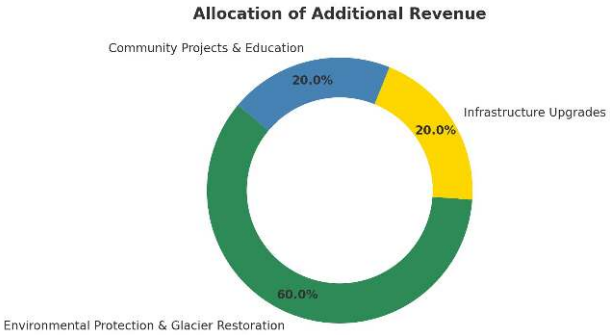
A Memo to the tourist council

To whom it may concern,

We present an overview of our sustainable tourism model for Juneau, designed to address overtourism by balancing economic growth, environmental sustainability, and social well-being.

Our model predicts that without intervention, peak-season daily visitors could surpass 30,000 by 2030, straining infrastructure, harming the environment, and reducing resident satisfaction. To counter these issues, we propose capping annual visitors at 1.3 million, limiting cruise ship arrivals to four per day, and redistributing demand from summer to spring and autumn. Additionally, implementing a $10 environmental fee, a 6% tax on high-impact activities, and a 10% hotel tax increase could generate $17 million annually to fund sustainability measures.





We propose allocating revenue strategically: 60% for environmental protection (glacier restoration, carbon offsets, waste management), 20% for infrastructure upgrades (roads, energy grids, public transit), and 20% for community development (housing, cultural preservation, resident programs).

Short-term (3 years): Limit visitor growth to under 6% annually and reduce environmental pressure (EPI) by 15% through conservation efforts.Medium-term (5 years): Increase tourism revenue by 10%, cut carbon emissions by 20%, and improve resident satisfaction (SPI) by redistributing visitors.Long-term (10 years): Establish Juneau as a model for sustainable tourism by harmonizing growth and preservation.

To ensure progress, we recommend quarterly policy reviews, promoting eco-tourism and off-season packages, and collaborating with cruise lines to enforce caps and encourage off-peak travel. These strategies will safeguard Juneau’ s natural beauty and social integrity while ensuring economic vitality.

Yours sincerely,

Team #2516029

References

[1] Tangen, B. A., & McDowell, J. (2016). Sustainable Tourism Development in Juneau, Alaska: Managing Growth in a Sensitive Environment. Journal of Sustainable Tourism, 24(6), 891-906.

[2] Gunderson, L. H., & Pritchard, L. (2017). Resilience and Adaptive Capacity in Juneau, Alaska's Tourism Economy: Strategies for Sustainable Development. Ecology and Society, 22(3), 15.

[3] Dwyer, L., & Kim, C. (2003). Destination competitiveness: Determinants and indicators. Current Issues in Tourism, 6(5), 369-414.

[4] Zhao, X., & Zhang, Y. (2017). Multi-objective optimization for sustainable tourism planning in mountainous regions: A case study in Zhangjiajie. Tourism Management, 59, 306-318.

[5] Gössling, S., & Hall, C. M. (2006). Tourism and global environmental change: Ecological, social, economic and political interrelationships. Routledge.

[6] Li, X. R., & Ryan, C. (2018). Sustainable tourism: A critical analysis of the global discourse. Tourism Recreation Research, 43(2), 160-174.